# Taoufik BENHSSEIN

## Embedded Electronics Engineer – Available Immediately

Maroc/France | +212767280528 | benhssein.taoufik@gmail.com | linkedin.com/in/taoufik-benhssein



## Profile

Electronics and Automation Systems Engineer with a Master's degree in Embedded Electronics and experience in electronic architecture modeling, embedded systems, and electronic circuit design. Experienced in MBSE methodologies, Java plugin development, and PCB design through engineering internships at Dassault Systèmes and APTIV. Currently seeking opportunities in embedded electronics, hardware design, and systems engineering.

## Technical Skills

**Embedded Systems**: STM32, ESP32, ARM Cortex-M, Microcontrollers, Embedded C/C++

**Electronics**: Analog & Digital Electronics, Electronic Circuit Design, Sensors, Power Electronics, PCB Design

**PCB Design**: Altium Designer, Schematic Capture, PCB Layout, Component Placement, Routing, DRC

**Communication Protocols**: UART, SPI, I²C, CAN, RS-232/485, Ethernet

**Modeling & Systems Engineering**: SysML, MBSE, CATIA Magic, IBM DOORS, Cameo Systems Modeler, Requirements Engineering, System Architecture

## Professional Experience

**Model Based Systems Engineering Intern – Dassault Systèmes** — Brest, France
*Software R&D Department – Electronics Branch* — Apr. 2025 – Sept. 2025

- Developed Java plugins for CATIA Magic to support MBSE-based electronic architecture modeling.
- Contributed to electronic architecture workflows and interoperability with external engineering platforms through API integration.
- Used GitLab for version control and CATIA Magic for functional testing and user experience evaluation.

**Electrical Engineering Intern – APTIV ( VERSIGENT )** — Tangier, Morocco
*Engineering Department* — Mar. 2024 – Jul. 2024

- Improvement and Optimization of Ultrasonic Welding Machines for the Automotive Industry.
- Design and implementation of an electronic circuit for uninterrupted power supply of machines based on an inverter and battery charger using Proteus 8 and Altium Designer.
- Replacement of HDD storage with non-volatile cloud-based flash memory.
- Predictive economic analysis and hardware assembly at the factory.

**Process Engineering Intern – YAZAKI Corporation** — Tangier, Morocco
*Process Department* — Jul. 2023 – Sept. 2023

- Design of an Intelligent Overhead Conveyor System for Cable Harness Transport.
- Feasibility study of the system.
- Automation and synchronization of the system using Automation Studio and LabVIEW.
- Assembly of the overhead conveyor system using CATIA V5.

## Academic Projects

Experimental Measurement Setup for an Analog Neural Network Based on Elastic Wave Propagation — Jan. 2025 – Apr. 2025
*Jean Lamour Laboratory, Acoustic research*

- Developed an experimental measurement setup to characterize an analog neural network based on elastic wave propagation.
- Configured piezoelectric transducers, function generator, and oscilloscope for excitation, measurement, and analysis of elastic-wave responses.
- Used MATLAB for experimental data acquisition/processing, signal analysis, visualization, and characterization of the network response.
- Performed experimental measurements to validate the behavior and performance of the proposed neural-network architecture.

Design and Testing of a High-Voltage DC Sensing Board for Electric Vehicles — Feb. 2024 – June 2024
*Mid-Semester Group Project, Altium Designer*

- Designed a high-voltage DC sensing PCB for monitoring and measuring high-voltage electrical systems in electric vehicles.
- Developed the schematic and PCB layout using Altium Designer, including component placement and routing.
- Considered electrical isolation, signal conditioning, and measurement accuracy for safe high-voltage monitoring.
- Performed PCB testing and validation to verify the board's electrical and measurement performance.

Facial Recognition System for Employee Attendance Management — Jan.2023 – Mar. 2023

*Python OpenCV & Excel*

- Developed a facial recognition system for automated employee identification and attendance tracking.
- Implemented face detection and recognition using OpenCV and image-processing techniques.
- Developed an automated attendance workflow to record and organize employee attendance data in Excel.
- Evaluated the system under different conditions to evaluate recognition reliability and attendance-recording accuracy.

## Education

Université de Lorraine — Nancy, France
*Master's Degree – Embedded Electronics Engineer* — 2024 – 2025
Main courses: System-Level SoC Modeling, Architectural Design of Digital and Mixed-Signal Circuits, Artificial Intelligence, Real-Time and Embedded Systems, Smart Sensors and Sensor Networks, Testability of Digital Integrated Circuits.

National School of Applied Sciences ENSA — Tangier, Morocco
*Engineering Degree – Electronics and Automation Systems Engineer* — 2019 – 2024
Main courses: Embedded Systems, Power Electronics, Robotics, Image processing.

## Languages

French: Professional
English: Professional (C1 ICIMS Certificate)
Arabic: Native

## Interests

Football
Interior Design
Gaming